



PREMIUM

Full-Length Mock Test

Mock 02

Civil Engineering

GATE CE

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Structural Engineering — Moment Distribution]

Max BM in simply supported beam with UDL is at:

- A) Supports
 - B) Midspan
 - C) Quarter
 - D) Anywhere
-

Q.2 [1 Mark] [EASY] [Geotechnical Engineering — Soil Properties]

Terzaghi's bearing capacity assumes:

- A) General shear
 - B) Local shear
 - C) Punching shear
 - D) None
-

Q.3 [1 Mark] [EASY] [Transportation Engineering — Traffic Engineering]

Bernoulli's equation is based on:

- A) Mass conservation
 - B) Energy conservation
 - C) Momentum conservation
 - D) Force conservation
-

Q.4 [1 Mark] [EASY] [Water Resources — Hydraulics]

Prandtl's mixing length theory is used for:

- A) Laminar flow
 - B) Turbulent flow
 - C) Compressible flow
 - D) Open channel
-

Q.5 [1 Mark] [EASY] [Environmental Engineering — Solid Waste]

CPM stands for:

- A) Critical Path Method
 - B) Cost Per Mile
 - C) Continuous Process Model
 - D) Critical Process Method
-

Q.6 [1 Mark] [EASY] [Construction Management — Valuation]

Max BM in simply supported beam with UDL is at:

- A) Supports
 - B) Midspan
 - C) Quarter
 - D) Anywhere
-

Q.7 [1 Mark] [EASY] [Structural Engineering — Deflection]

Terzaghi's bearing capacity assumes:

A) General shear

B) Local shear

C) Punching shear

D) None

Q.8 [1 Mark] [EASY] [Geotechnical Engineering — Slope Stability]

Bernoulli's equation is based on:

- A) Mass conservation
 - B) Energy conservation
 - C) Momentum conservation
 - D) Force conservation
-

Q.9 [1 Mark] [EASY] [Transportation Engineering — Traffic Engineering]

Prandtl's mixing length theory is used for:

- A) Laminar flow
 - B) Turbulent flow
 - C) Compressible flow
 - D) Open channel
-

Q.10 [1 Mark] [EASY] [Water Resources — Hydrology]

CPM stands for:

- A) Critical Path Method
 - B) Cost Per Mile
 - C) Continuous Process Model
 - D) Critical Process Method
-

Q.11 [1 Mark] [EASY] [Environmental Engineering — Sewage Treatment]

Max BM in simply supported beam with UDL is at:

- A) Supports
 - B) Midspan
 - C) Quarter
 - D) Anywhere
-

Q.12 [1 Mark] [EASY] [Construction Management — Project Planning]

Terzaghi's bearing capacity assumes:

- A) General shear
 - B) Local shear
 - C) Punching shear
 - D) None
-

Q.13 [2 Marks] [MODERATE] [Structural Engineering — Influence Lines]

Bernoulli's equation is based on:

- A) Mass conservation
 - B) Energy conservation
 - C) Momentum conservation
 - D) Force conservation
-

Q.14 [2 Marks] [MODERATE] [Geotechnical Engineering — Permeability]

Prandtl's mixing length theory is used for:

- A) Laminar flow

B) Turbulent flow

C) Compressible flow

D) Open channel

Q.15 [2 Marks] [MODERATE] [Transportation Engineering — Railways]

CPM stands for:

- A) Critical Path Method
 - B) Cost Per Mile
 - C) Continuous Process Model
 - D) Critical Process Method
-

Q.16 [2 Marks] [MODERATE] [Water Resources — Irrigation]

Max BM in simply supported beam with UDL is at:

- A) Supports
 - B) Midspan
 - C) Quarter
 - D) Anywhere
-

Q.17 [2 Marks] [MODERATE] [Environmental Engineering — Water Treatment]

Terzaghi's bearing capacity assumes:

- A) General shear
 - B) Local shear
 - C) Punching shear
 - D) None
-

Q.18 [2 Marks] [MODERATE] [Construction Management — Contracts]

Bernoulli's equation is based on:

- A) Mass conservation
 - B) Energy conservation
 - C) Momentum conservation
 - D) Force conservation
-

Q.19 [2 Marks] [MODERATE] [Structural Engineering — Slope Deflection]

Prandtl's mixing length theory is used for:

- A) Laminar flow
 - B) Turbulent flow
 - C) Compressible flow
 - D) Open channel
-

Q.20 [2 Marks] [MODERATE] [Geotechnical Engineering — Soil Properties]

CPM stands for:

- A) Critical Path Method
 - B) Cost Per Mile
 - C) Continuous Process Model
 - D) Critical Process Method
-

Q.21 [2 Marks] [MODERATE] [Transportation Engineering — Railways]

Max BM in simply supported beam with UDL is at:

- A) Supports

B) Midspan

C) Quarter

D) Anywhere

Q.22 [2 Marks] [MODERATE] [Water Resources — Hydraulics]

Terzaghi's bearing capacity assumes:

- A) General shear
- B) Local shear
- C) Punching shear
- D) None

Q.23 [2 Marks] [MODERATE] [Environmental Engineering — Water Treatment]

Bernoulli's equation is based on:

- A) Mass conservation
- B) Energy conservation
- C) Momentum conservation
- D) Force conservation

Q.24 [2 Marks] [MODERATE] [Construction Management — Contracts]

Prandtl's mixing length theory is used for:

- A) Laminar flow
- B) Turbulent flow
- C) Compressible flow
- D) Open channel

Q.25 [2 Marks] [MODERATE] [Structural Engineering — Moment Distribution]

CPM stands for:

- A) Critical Path Method
- B) Cost Per Mile
- C) Continuous Process Model
- D) Critical Process Method

Q.26 [2 Marks] [MODERATE] [Geotechnical Engineering — Bearing Capacity]

Max BM in simply supported beam with UDL is at:

- A) Supports
- B) Midspan
- C) Quarter
- D) Anywhere

Q.27 [2 Marks] [MODERATE] [Transportation Engineering — Railways]

Terzaghi's bearing capacity assumes:

- A) General shear
- B) Local shear
- C) Punching shear
- D) None

Q.28 [2 Marks] [MODERATE] [Water Resources — Open Channel Flow]

Bernoulli's equation is based on:

- A) Mass conservation

B) Energy conservation

C) Momentum conservation

D) Force conservation

Q.29 [2 Marks] [MODERATE] [Environmental Engineering — Noise Pollution]

Prandtl's mixing length theory is used for:

- A) Laminar flow
 - B) Turbulent flow
 - C) Compressible flow
 - D) Open channel
-

Q.30 [2 Marks] [MODERATE] [Construction Management — Valuation]

CPM stands for:

- A) Critical Path Method
 - B) Cost Per Mile
 - C) Continuous Process Model
 - D) Critical Process Method
-

Q.31 [2 Marks] [MODERATE] [Structural Engineering — Influence Lines]

Max BM in simply supported beam with UDL is at:

- A) Supports
 - B) Midspan
 - C) Quarter
 - D) Anywhere
-

Q.32 [2 Marks] [MODERATE] [Geotechnical Engineering — Soil Properties]

Terzaghi's bearing capacity assumes:

- A) General shear
 - B) Local shear
 - C) Punching shear
 - D) None
-

Q.33 [2 Marks] [HARD] [Transportation Engineering — Highway Planning]

Bernoulli's equation is based on:

- A) Mass conservation
 - B) Energy conservation
 - C) Momentum conservation
 - D) Force conservation
-

Q.34 [2 Marks] [HARD] [Water Resources — Open Channel Flow]

Prandtl's mixing length theory is used for:

- A) Laminar flow
 - B) Turbulent flow
 - C) Compressible flow
 - D) Open channel
-

Q.35 [2 Marks] [HARD] [Environmental Engineering — Noise Pollution]

CPM stands for:

- A) Critical Path Method

B) Cost Per Mile

C) Continuous Process Model

D) Critical Process Method

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Construction Management — Valuation]

Which are statically determinate?

- A) Simply supported
 - B) Fixed-fixed
 - C) Cantilever
 - D) Continuous
 - E) Three-hinged arch
-

Q.37 [2 Marks] [HARD] [Structural Engineering — Slope Deflection]

Which are statically determinate?

- A) Simply supported
 - B) Fixed-fixed
 - C) Cantilever
 - D) Continuous
 - E) Three-hinged arch
-

Q.38 [2 Marks] [HARD] [Geotechnical Engineering — Slope Stability]

Which are statically determinate?

- A) Simply supported
 - B) Fixed-fixed
 - C) Cantilever
 - D) Continuous
 - E) Three-hinged arch
-

Q.39 [2 Marks] [HARD] [Transportation Engineering — Traffic Engineering]

Which are statically determinate?

- A) Simply supported
 - B) Fixed-fixed
 - C) Cantilever
 - D) Continuous
 - E) Three-hinged arch
-

Q.40 [2 Marks] [HARD] [Water Resources — Hydraulics]

Which are statically determinate?

- A) Simply supported
 - B) Fixed-fixed
 - C) Cantilever
 - D) Continuous
 - E) Three-hinged arch
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Transportation Engineering — Pavement Design]**

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: _____

Q.52 [2 Marks] [HARD] [Water Resources — Hydrology]

Sanitary landfill compaction target: density (kg/m^3)?

Answer: _____

Q.53 [2 Marks] [HARD] [Environmental Engineering — Sewage Treatment]

SS beam span 6m, UDL 10kN/m. Max BM ($\text{kN}\cdot\text{m}$)?

Answer: _____

Q.54 [2 Marks] [HARD] [Construction Management — Valuation]

Rod $A=500\text{mm}^2$, $P=50\text{kN}$, $E=200\text{GPa}$, $L=2\text{m}$. Elongation (mm)?

Answer: _____

Q.55 [2 Marks] [HARD] [Structural Engineering — Moment Distribution]

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: _____

Q.56 [2 Marks] [HARD] [Geotechnical Engineering — Consolidation]

Sanitary landfill compaction target: density (kg/m^3)?

Answer: _____

Q.57 [2 Marks] [HARD] [Transportation Engineering — Pavement Design]

SS beam span 6m, UDL 10kN/m. Max BM ($\text{kN}\cdot\text{m}$)?

Answer: _____

Q.58 [2 Marks] [HARD] [Water Resources — Hydraulics]

Rod $A=500\text{mm}^2$, $P=50\text{kN}$, $E=200\text{GPa}$, $L=2\text{m}$. Elongation (mm)?

Answer: _____

Q.59 [2 Marks] [HARD] [Environmental Engineering — Solid Waste]

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: _____

Q.60 [2 Marks] [HARD] [Construction Management — Quality Control]

Sanitary landfill compaction target: density (kg/m^3)?

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	B	Q.2:	A	Q.3:	B
Q.4:	B	Q.5:	A	Q.6:	B
Q.7:	A	Q.8:	B	Q.9:	B
Q.10:	A	Q.11:	B	Q.12:	A
Q.13:	B	Q.14:	B	Q.15:	A
Q.16:	B	Q.17:	A	Q.18:	B
Q.19:	B	Q.20:	A	Q.21:	B
Q.22:	A	Q.23:	B	Q.24:	B
Q.25:	A	Q.26:	B	Q.27:	A
Q.28:	B	Q.29:	B	Q.30:	A
Q.31:	B	Q.32:	A	Q.33:	B
Q.34:	B	Q.35:	A	Q.36:	ACE
Q.37:	ACE	Q.38:	ACE	Q.39:	ACE
Q.40:	ACE	Q.41:	ACE	Q.42:	ACE
Q.43:	ACE	Q.44:	ACE	Q.45:	ACE
Q.46:	ACE	Q.47:	ACE	Q.48:	ACE
Q.49:	ACE	Q.50:	ACE	Q.51:	400000
Q.52:	600	Q.53:	45	Q.54:	1
Q.55:	400000	Q.56:	600	Q.57:	45
Q.58:	1	Q.59:	400000	Q.60:	600

Detailed Solutions

Question 1 [Structural Engineering — Moment Distribution]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 2 [Geotechnical Engineering — Soil Properties]

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 3 [Transportation Engineering — Traffic Engineering]

Bernoulli's equation is based on:

Answer: (B)

$P/\rho + v^2/2g + z = \text{constant}$. Conservation of energy.

Question 4 [Water Resources — Hydraulics]

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 5 [Environmental Engineering — Solid Waste]

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 6 [Construction Management — Valuation]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 7 [Structural Engineering — Deflection]

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 8 [Geotechnical Engineering — Slope Stability]

Bernoulli's equation is based on:

Answer: (B)

$P/\rho + v^2/2g + z = \text{constant}$. Conservation of energy.

Question 9 [Transportation Engineering — Traffic Engineering]

Prandtl's mixing length theory is used for:

Answer:

(B)

Mixing length theory models turbulent shear stress.

Question 10 [Water Resources — Hydrology]

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 11 [Environmental Engineering — Sewage Treatment]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 12 [Construction Management — Project Planning]

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 13 [Structural Engineering — Influence Lines]

Bernoulli's equation is based on:

Answer: (B)

$P/\rho + \frac{1}{2}V^2 + gz = \text{constant}$. Conservation of energy.

Question 14 [Geotechnical Engineering — Permeability]

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 15 [Transportation Engineering — Railways]

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 16 [Water Resources — Irrigation]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 17 [Environmental Engineering — Water Treatment]

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 18 [Construction Management — Contracts]

Bernoulli's equation is based on:

Answer:

(B)

$P < r^2 V^2/2g + z = \text{constant}$. Conservation of energy.

Question 19 [Structural Engineering — Slope Deflection]

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 20 [Geotechnical Engineering — Soil Properties]

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 21 [Transportation Engineering — Railways]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 22 [Water Resources — Hydraulics]

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 23 [Environmental Engineering — Water Treatment]

Bernoulli's equation is based on:

Answer: (B)

$P < r^2 V^2/2g + z = \text{constant}$. Conservation of energy.

Question 24 [Construction Management — Contracts]

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 25 [Structural Engineering — Moment Distribution]

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 26 [Geotechnical Engineering — Bearing Capacity]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 27 [Transportation Engineering — Railways]

Terzaghi's bearing capacity assumes:

Answer:

Terzaghi theory based on general shear failure mode.

Question 28 [Water Resources — Open Channel Flow]

Bernoulli's equation is based on:

Answer: (B)

$P/\rho + V^2/2g + z = \text{constant}$. Conservation of energy.

Question 29 [Environmental Engineering — Noise Pollution]

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 30 [Construction Management — Valuation]

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 31 [Structural Engineering — Influence Lines]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 32 [Geotechnical Engineering — Soil Properties]

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 33 [Transportation Engineering — Highway Planning]

Bernoulli's equation is based on:

Answer: (B)

$P/\rho + V^2/2g + z = \text{constant}$. Conservation of energy.

Question 34 [Water Resources — Open Channel Flow]

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 35 [Environmental Engineering — Noise Pollution]

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 36 [Construction Management — Valuation]

Which are statically determinate?

Answer:

(A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 37 [Structural Engineering — Slope Deflection]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 38 [Geotechnical Engineering — Slope Stability]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 39 [Transportation Engineering — Traffic Engineering]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 40 [Water Resources — Hydraulics]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 41 [Environmental Engineering — Solid Waste]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 42 [Construction Management — Project Planning]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 43 [Structural Engineering — Influence Lines]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 44 [Geotechnical Engineering — Consolidation]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 45 [Transportation Engineering — Airport Planning]

Which are statically determinate?

Answer:

(A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 46 [Water Resources — Open Channel Flow]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 47 [Environmental Engineering — Solid Waste]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 48 [Construction Management — Contracts]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 49 [Structural Engineering — Slope Deflection]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 50 [Geotechnical Engineering — Consolidation]

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 51 [Transportation Engineering — Pavement Design]

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: (400000)

$Re = VD/\nu$ (where ν is kinematic viscosity).

Question 52 [Water Resources — Hydrology]

Sanitary landfill compaction target: density (kg/m^3)?

Answer: (600)

Typical compacted MSW density is 600 kg/m^3 .

Question 53 [Environmental Engineering — Sewage Treatment]

SS beam span 6m, UDL 10kN/m. Max BM ($\text{kN}\cdot\text{m}$)?

Answer: (45)

Max BM = $wL^2/8 = 10 \times 36/8 = 45 \text{ kN}\cdot\text{m}$.

Question 54 [Construction Management — Valuation]

Rod A=500mm², P=50kN, E=200GPa, L=2m. Elongation (mm)?

Answer:

;B Ò Â R Ò S s"òfS s²ms# s " Ò ã Ò Ò ÖÖà

Question 55 [Structural Engineering — Moment Distribution]

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: (400000)

$Re = VD/\nu$; Ò Ò -s ã"ó s² b Ò C ±GW& ulent).

Question 56 [Geotechnical Engineering — Consolidation]

Sanitary landfill compaction target: density (kg/m³)?

Answer: (600)

Typical compacted MSW density "H 600 kg/m³.

Question 57 [Transportation Engineering — Pavement Design]

SS beam span 6m, UDL 10kN/m. Max BM (kN·m)?

Answer: (45)

Max BM = $wL^2/8 = 10 \times 36/8 = 45$ kN·m.

Question 58 [Water Resources — Hydraulics]

Rod A=500mm², P=50kN, E=200GPa, L=2m. Elongation (mm)?

Answer: (1)

;B Ò Â R Ò S s"òfS s²ms# s " Ò ã Ò Ò ÖÖà

Question 59 [Environmental Engineering — Solid Waste]

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: (400000)

$Re = VD/\nu$; Ò Ò -s ã"ó s² b Ò C ±GW& ulent).

Question 60 [Construction Management — Quality Control]

Sanitary landfill compaction target: density (kg/m³)?

Answer: (600)

Typical compacted MSW density "H 600 kg/m³.

